4.6 Integrals With No Formula

WORKSHEET

The question

STEP III 2016, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q3

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself If $\int \mathrm{f}(x)e^x\,\mathrm{d}x = \frac{\mathrm{P}}{\mathrm{Q}}e^x + C$, what equation must P and Q satisfy? (Differentiate the right-hand side.)

STAGE 1 The factor $x+1$

Covers: Part (i)

Ask yourself Differentiate $\frac{\mathrm{P}}{\mathrm{Q}}e^x$ and set it equal to $\mathrm{f}(x)e^x$. Clear the denominators to get an identity between polynomials.

Pause and try Put $x=-1$ in that identity. What must be true of $\mathrm{Q}(-1)$?

STAGE 2 Degrees

Covers: Part (i)

Pause and try Let P and Q have degrees p and q. In $\mathrm{P}'\mathrm{Q} - \mathrm{P}\mathrm{Q}' + \mathrm{P}\mathrm{Q}$, which term has the highest degree? Compare the degrees of the two sides.

STAGE 3 Finding P

Covers: Part (i)

Ask yourself With $\mathrm{Q} = x + 1$, P is a quadratic. Write $\mathrm{P} = \alpha x^2 + \beta x + \gamma$, substitute, and compare coefficients.

STAGE 4 An impossibility

Covers: Part (ii)

Ask yourself Write the identity for $f(x) = \frac{1}{x+1}$. As in part (i), show that $Q(-1) = 0$.

Pause and try Write $Q = (x+1)^k R$ with $k \geq 1$ and $R(-1) \neq 0$. Substitute, cancel as many factors of $x+1$ as you can, and put $x = -1$. What do you learn about $P(-1)$?

What's really going on

Poles and logarithms

There is a simple picture behind part (ii). The integrand $\frac{e^x}{x+1}$ has a *simple pole* at $x = -1$: near there it behaves like $\frac{e^{-1}}{x+1}$. Integrating $\frac{1}{x+1}$ gives $\ln|x+1|$, so the antiderivative should behave like $e^{-1}\ln|x+1|$ near -1, growing to $-\infty$ logarithmically (Figure 4.6.1). But a rational function times e^x can only blow up like a power $(x+1)^{-k}$, and its derivative then has a pole of order $k+1 \geq 2$, never a simple pole. Logarithmic behaviour cannot be produced by rational functions; the algebra in Stage 4 is a rigorous version of this observation. In part (i), by contrast, the integrand has a *double* pole, which is exactly what the derivative of something with a simple pole looks like, so a rational answer was possible.

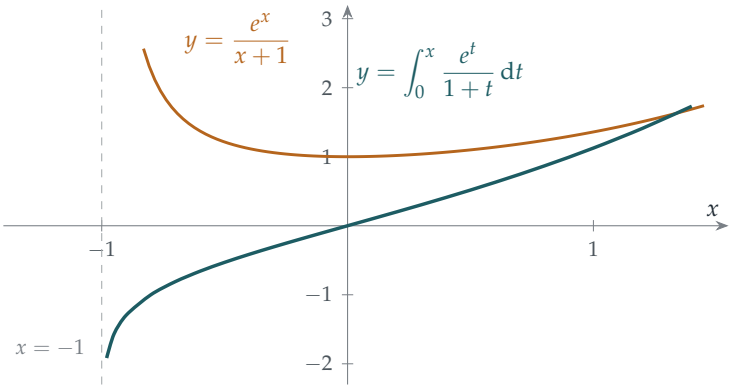


Figure 4.6.1 The integrand $\frac{e^x}{x+1}$ (amber) has a simple pole at $x = -1$. Its antiderivative (teal, computed numerically) exists perfectly well, but it tends to $-\infty$ *logarithmically* as $x \to -1^+$, like $e^{-1}\ln(x+1)$. No rational function times e^x can behave like that.

Liouville and the Risch algorithm

In the 1830s Joseph Liouville developed a theory of which integrals can be expressed in terms of *elementary functions* (those built from polynomials, exponentials, logarithms, roots and trigonometric functions by the usual operations). One of his results says that $\int f(x)e^{g(x)}\,dx$, with f and g rational and g not constant, is elementary only if it equals $R(x)e^{g(x)}$ (plus a constant) for some rational R. (The condition on g matters: with $g = 0$ and $f(x) = \frac{1}{x}$ the integral is $\ln|x|$, which is elementary but not rational.) So this question, which rules out the rational form, actually proves that $\int \frac{e^x}{x+1}\,dx$ is not elementary at all. Versions of the same theorem show that $\int e^{-x^2}\,dx$ and $\int \frac{\sin x}{x}\,dx$ are not elementary. In 1969 Robert Risch turned Liouville's ideas into an algorithm that decides, for a large class of functions, whether an elementary integral exists and finds it if it does. Versions of it run inside computer algebra systems.

Special functions

When an integral has no elementary form, mathematicians give it a name and study it as a new function, just as we did with the Lambert W function in Section 3.9. The integral here is essentially the *exponential integral* $\mathrm{Ei}(x)$, which appears in heat flow and in the *logarithmic integral* $\mathrm{li}(x) = \mathrm{Ei}(\ln x)$, the best simple approximation to the number of primes up to x (Section 1.2). For $x < 0$ it is $\int_{-\infty}^{x} \frac{e^t}{t}\,dt$. For $x > 0$ that integral runs through the infinite spike at $t = 0$, so $\mathrm{Ei}(x)$ is defined by cutting out a small interval $(-\varepsilon, \varepsilon)$ around the spike and letting $\varepsilon \to 0$: the *Cauchy principal value*.

Try this next

1 Two that work. Show that $\displaystyle\int \frac{xe^x}{(x+1)^2}\,dx = \frac{e^x}{x+1} + C$ and $\displaystyle\int \frac{(x^2+1)e^x}{(x+1)^2}\,dx = \frac{x-1}{x+1}\,e^x + C$.

2 The Gaussian. Show that there is no polynomial P with $\displaystyle\int e^{-x^2}\,dx = P(x)e^{-x^2} + C$. (*Hint:* differentiate and compare degrees.)

3 Another pole. Adapt the argument of part (ii) to show that $\displaystyle\int \frac{e^x}{x}\,dx$ is not of the form $\frac{P(x)}{Q(x)}e^x + C$.

4 When is it possible? Suppose f is a polynomial and $\displaystyle\int \frac{f(x)}{(x+1)^2}\,e^x\,dx = \frac{P(x)}{x+1}\,e^x + C$ for some polynomial P. Show that $(x+1)P' + xP = f$, and by evaluating this and its derivative at $x = -1$, show that $f(-1) + f'(-1) = 0$. Check that this condition holds for $f(x) = x^3 - 2$ and for the examples in exercise 1. HARDERCan you show that the condition is also sufficient?

Further reading

- **Brian Conrad, "Impossibility theorems for elementary integration"**. Freely available lecture notes explaining Liouville's theorem with examples.
- **Elena Anne Marchisotto & Gholam-Ali Zakeri, "An invitation to integration in finite terms"**, *The College Mathematics Journal* 25 (1994). A gentle introduction to Liouville's theory.
- **Manuel Bronstein, *Symbolic Integration I*** (Springer). How computer algebra systems actually integrate. Advanced.

The full guided solution is in Section 4.6:
https://beyondstep.keithmcnulty.org/ch04/no-formula.html